



Bridging Speech and Singing: Multi-stage Speech-Prompted Singing Voice Conversion with Speaker Embedding Adaptation

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Abstract

Singing voice conversion (SVC) technology holds immense potential to transform the entertainment industry and advance human-computer interaction. Unlike previous SVC studies that primarily focus on the conversion between different singing voices, this work specifically investigates the conversion of speech timbre into singing. In this paper, we introduce a novel Singing Speech Alignment Network (SSAN) designed to align speaker and singer embeddings derived from input data. To effectively train the SVC model, we also propose a cycle training strategy that ensures the performance of SVC with speech prompts. Experimental results demonstrate that the proposed method (SSANSVC) achieves superior performance in both the naturalness of the synthesized singing and its similarity to the target speech.

Index Terms: Singing Voice Conversion, Speaker-singer Adaptation, Cycle Training

1. Introduction

Singing Voice Conversion (SVC) aims to transform the timbre of a song's singer into that of another singer while preserving the content and melody of the singing. It not only holds tremendous potential in the entertainment industry and content creation but also shows broad application prospects in fields such as education and personalized voice generation [1].

SVC models are trained using either parallel or non-parallel datasets. The parallel dataset requires multiple singers performing the same song, while the non-parallel dataset is more flexible to involve multiple singers performing different songs. Early efforts in SVC focus more on using parallel data. These methods build the mapping of the parallel samples using statistical techniques [2, 3].

However, the scale of parallel datasets is limited due to strict requirements, which has led to increasing attention on methods that use non-parallel data for SVC. An unsupervised deep learning SVC method based on WaveNet [4] is proposed, incorporating a singer-agnostic latent representation embedding to enhance the flexibility of the conversion [5]. To address the issue of converted singing voices often going out of tune, a pitch adversarial mechanism is proposed, which forces the encoder to learn pitch-invariant phoneme representations from the source audio. Then, the extracted pitch information is further fed into the decoder network, significantly improving the quality of the converted singing voice [6]. In addition, Variational Auto-Encoders (VAEs) [7] and Variational Autoencoding Wasserstein Generative Adversarial Networks (VAW-GANs) [8] have also been explored for non-parallel SVC.

Recently, diffusion models have emerged as a powerful generative technique in the fields of image and audio synthe-

sis [9–13]. These models learn the data distribution by gradually introducing and removing noise to capture subtle features in audio signals. Due to its remarkable ability to model complex data distributions, it has become a promising alternative for SVC tasks [14, 15]. A method based on a denoising diffusion probabilistic model, DiffSVC, is proposed for singing voice conversion. By using PPGs, fundamental frequency features, and loudness features as auxiliary inputs, it significantly improves the naturalness and timbre similarity of the converted singing voice [14]. The performance of diffusion model-based SVC methods is outstanding. However, the iterative sampling process leads to slow inference speed. To mitigate the challenge, a consistency model-based SVC method, CoMoSVC, is proposed to achieve high-speed sampling [15].

However, previous SVC research has generally focused on the conversion between different singers' voices, with little attention given to the conversion of the speech of a target speaker to singing.¹ Speech signals emphasize clarity and intelligibility, while singing is usually more diverse, often incorporating richer emotional expression. Therefore, converting speech prompts to singing requires careful modeling of the intricate details of timbre. Additionally, datasets containing both speech and singing from the same speaker are relatively scarce, which presents an even greater challenge for research on speech-prompted SVC.

According to the above challenges in speech-prompted SVC, we design a speaker-singer adaptation network (SSAN) to bridge the gap between speaker embeddings (extracted from speech) and singer embeddings (extracted from singing), thereby enhancing the quality of timbre conversion from speech to singing. Furthermore, we adapted a multi-stage cycle training strategy to further refine the performance of the proposed method, namely SSANSVC. Experimental results demonstrate that our approach outperforms baseline models in both objective evaluations and subjective metrics. The main contributions of this work can be summarized as follows:

1. We design an SSAN to align speaker and singer embeddings from input audio clips. This network is specifically tailored to bridge the inherent differences between speech and singing data by capturing and adapting the unique characteristics of each domain.

2. To address the issue of the scarcity of paired speech data and singing data from the same speaker/singer, we propose a multi-stage cycle training strategy to mitigate this challenge. By adopting the strategy, we successfully achieve cross-domain singing voice conversion, effectively utilizing the available unpaired data.

¹Different from previous speech-to-singing conversion work [16–18], which focuses on transferring speech to singing voice, this paper emphasizes timbre conversion between speech and singing.

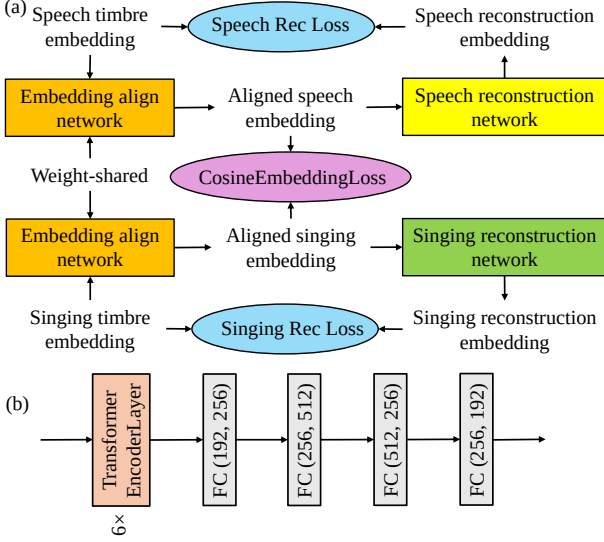


Figure 1: The framework of the SSAN.

2. Method

In this section, we first describe the SSAN to improve the speaker embedding network for better transferability in the SVC modeling. Next, we present the overall architecture of the SVC model. Finally, we outline the proposed cycle training strategy to enhance the SVC performance and improve timbre disentanglement.

2.1. Speaker-Singer Adaptation Network

As illustrated in Figure 1(a), the framework of the proposed SSAN is composed of three modules, namely Embedding align network, Speech reconstruction network, and Singing reconstruction network. The architecture of the above three layers is the same, as shown in Figure 1(b).

The Embedding align network aligns the timbre embeddings of speech and singing signals. It captures similarities within the same speaker or singer while distinguishing differences between different ones. This helps preserve the speaker’s unique characteristics while handling stylistic variations in singing. The Speech reconstruction network and Singing reconstruction network are responsible for restoring timbre embeddings of the speech and singing signals processed by the Embedding align network to their original feature representations. This design helps prevent overfitting of the network, ensuring that it does not deviate from the original characteristics during training, thereby maintaining the key information of the original data.

During training, speech timbre embedding t_{speech} and singing timbre embedding t_{singing} are first passed through the Embedding align network, where t_{speech} and t_{singing} are extracted by ESPnet-SPK [19].² One sample embedding is selected from each of the two datasets to form a speaker-singer timbre embedding pair $\{t_{\text{speech}}, t_{\text{singing}}\}$. We assign -1 as target for mismatched speaker-singer timbre embedding pairs and 1 as target for matched timbre embedding pairs. Then, the timbre embedding differences between the two training samples are computed, and the predicted similarity value is obtained through the similarity adaptation layer. Afterward, the aligned t_{speech}

and t_{singing} processed by the Embedding align network are fed into the Speech reconstruction network and the Singing reconstruction network, where the aligned t_{speech} and t_{singing} are reconstructed to ensure that the timbre embeddings are restored to match the original embeddings. The specific training process of a batch is described in Algorithm 1. The loss of training SSAN can be expressed as:

$$L = L_{\text{cosineEmbeddingLoss}} + L_{\text{speech Rec}} + L_{\text{singing Rec}} \quad (1)$$

$$L_{\text{cosineEmbeddingLoss}} = \begin{cases} 1 - \frac{t_{\text{speech}} \cdot t_{\text{singing}}}{\|t_{\text{speech}}\| \cdot \|t_{\text{singing}}\|}, & \text{if } y = 1 \\ \max\left(0, \frac{t_{\text{speech}} \cdot t_{\text{singing}}}{\|t_{\text{speech}}\| \cdot \|t_{\text{singing}}\|} + 1\right), & \text{if } y = -1 \end{cases} \quad (2)$$

where $L_{\text{speech Rec}} = \|t_{\text{speech}} - \hat{t}_{\text{speech}}\|_2$ and $L_{\text{singing Rec}} = \|t_{\text{singing}} - \hat{t}_{\text{singing}}\|_2$ are the reconstruction loss, $\|\cdot\|_2$ is the L_2 norm distance, y is the target label, as 1 for matching, and as -1 for mismatching.

Algorithm 1 The pseudocode for the SSAN.

Input: Training set V_{speech} and V_{singing}

Output: Predicted similarity distance $\hat{y}$

- 1: Randomly select $\{t_{\text{speech}}^{(1)}, \dots, t_{\text{speech}}^{(M)}\}$ from V_{speech} ;
- 2: Randomly generate labels $\{y_1, \dots, y_M\} \in \{-1, 1\}$;
- 3: **for** $k = 1$ to M **do**
- 4: **if** $y_k = -1$ **then**
- 5: Choose $t_{\text{singing}}^{(k)}$ from V_{singing} with same speaker as $t_{\text{speech}}^{(k)}$;
- 6: **else**
- 7: Choose $t_{\text{singing}}^{(k)}$ from V_{singing} with different speaker from $t_{\text{speech}}^{(k)}$;
- 8: **end if**
- 9: Construct support data $s_k = \{t_{\text{speech}}^{(k)}, t_{\text{singing}}^{(k)}; y_k\}$;
- 10: **end for**
- 11: Compute loss L and update network parameters;
- 12: **return** Trained model SSAN;

2.2. SVC Model Architecture

Our method is based on the CoMoSVC teacher architecture [15], where the main difference is to replace the traditional singer-id with the speaker timbre feature extracted from the SSAN model proposed in Sec. 2.1. The proposed method is named SSANSVC. Different from CoMoSVC, which requires training on each specific target domain, SSANSVC enables zero-shot capability, allowing the model to generalize to unseen target timbres without requiring additional training. Similar to CoMoSVC, the SSANSVC model consists of two parts, encoding and decoding modules. For the encoding process, the model encodes the timbre-related information and timbre-independent features extracted from the signal (shown as $\mathbf{S}_{\{\text{Speech}, \text{Sing}\}}$ in Figure 2) into embedded feature representations, which are subsequently concatenated and used as conditional inputs that are passed to the decoding module to generate the mel-spectrogram. Next, the generated mel-spectrogram is converted into an audio signal using a pre-trained vocoder.

Specifically, singing signals contain many components mixed together such as content, pitch, loudness, and singer timbre. During the encoding process, timbre-related and timbre-independent features are extracted. Timbre features are obtained by the SSAN described in Sec. 2.1, shown as $t_{\{\text{Speech}, \text{Sing}\}}$

²https://huggingface.co/espnet/voxcelebs12_rawnet3

in Figure 2. Content features are extracted by using a pre-trained acoustic model ContentVec [20]. Pitch information is estimated using the widely used classical F0 estimator DIO [21]. Loudness information is derived from the squared amplitude of the signal. The content features, pitch information and loudness information are shown as $c_{\{\text{Speech, Sing}\}}$ in Figure 2. After feature extraction, a linear layer is applied to all embeddings to standardize their dimensions, and they are concatenated to form the conditional inputs for the decoding module. The decoding process remains the same as in [15].

2.3. Cycle Training

As discussed in Sec. 1, obtaining speech and singing data from the same speaker is both expensive and challenging. A common approach to this challenge is to combine speech and singing data from different speakers/singers to create a mixed dataset with speech and singing data. Content and timbre information are then extracted from each dataset and used to reconstruct audio data. However, since the content and timbre feature during training are from the same speaker’s data, existing models struggle to generate data where the content is derived from singing data and the timbre from speech data. Inspired from [22], we propose a cycle training strategy to address this limitation and fully exploit the potential of a dataset that has both speech and singing voices.

As depicted in Figure 2, the proposed cycle training strategy consists of two stages. Stage 1 is similar to the previous works in SVC [14, 15], where the model takes the data of the same person as both content input and timbre input to reconstruct the data. For stage 2, the source audio for content input and the source audio for timbre input are essentially different types of data (i.e., speech or singing), simulating a singing voice conversion with the speech-prompted scenario. We randomly select two speakers for this illustration: speaker 1, who speaks, and speaker 2, who sings. We represent a speech segment from speaker 1 as $\mathbf{S}_{\text{Speech}}^{(1)-(a)}$, and another speech segment from speaker 1 as $\mathbf{S}_{\text{Speech}}^{(1)-(c)}$, where (1) refers to speaker identity, and (a), (c) refers to two distinct utterances. Similarly, singing data spoken by speaker 2 is present as $\mathbf{S}_{\text{Singing}}^{(2)-(b)}$.

Specifically, we employ $\mathbf{S}_{\text{Singing}}^{(2)-(b)}$ of speaker 2 from singing data as content input and take $\mathbf{S}_{\text{Speech}}^{(1)-(a)}$ of speaker 1 from speech data as timbre input in stage 2, and use the model to generate a singing segment $\hat{\mathbf{S}}_{\text{Singing}}^{(1)-(b)}$ which assume the content information is the same as $\mathbf{S}_{\text{Singing}}^{(2)-(b)}$ but with the timbre of speaker 1. To formulate a cycle training pipeline, we then utilize the output $\hat{\mathbf{S}}_{\text{Singing}}^{(1)-(b)}$ as the timbre input. As for the content input, we use another speech sample from speaker 1, denoted as $\mathbf{S}_{\text{Speech}}^{(1)-(c)}$ in Figure 2. Since both speech segments possess the timbre of speaker 1, we simulated the speech segment and singing segment belonging to the same speaker. Consequently, another SVC pipeline with speech as a prompt can be performed. By incorporating stage 1 and the cross-type SVC cycle of stage 2, we extend the model’s ability in cross-type SVC with speech as the prompt.

3. Experiment

In this section, we first introduce the corpus used for the experiment and the data preprocessing methods. Then, we describe the implementation details of the baseline methods and the proposed method. Finally, we present the evaluation results.

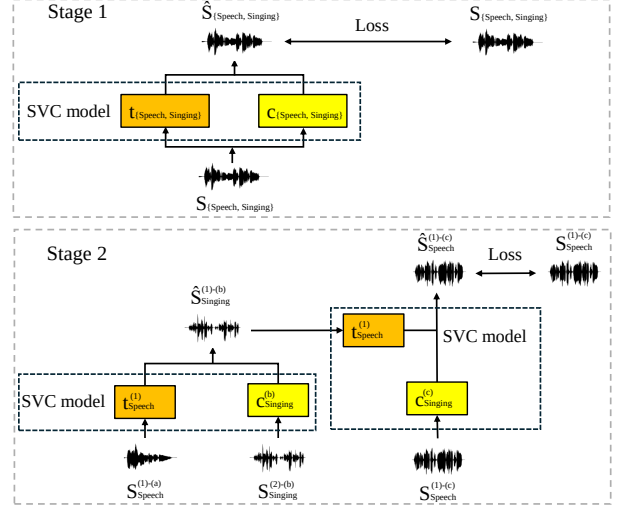


Figure 2: The flow chart of the cycle training strategy. In stage 1, the features $t_{\{\text{Speech, Sing}\}}$ and $c_{\{\text{Speech, Sing}\}}$ are extracted from data of the same type, meaning that speech features are derived from speech data, and singing features are derived from singing data. In stage 2, $\mathbf{S}_{\text{Speech}}^{(1)-(a)}$, $\mathbf{S}_{\text{Speech}}^{(1)-(c)}$ and $\hat{\mathbf{S}}_{\text{Singing}}^{(1)-(b)}$ represent speech data, $\mathbf{S}_{\text{Singing}}^{(2)-(b)}$ and $\hat{\mathbf{S}}_{\text{Singing}}^{(1)-(b)}$ represent singing data, where (1)(2) denote different speaker identities, and (a)(b)(c) denote different contents. Please refer to Sec. 2.3 for more discussion.

3.1. Experimental Setup

Dataset: Several datasets are adopted to train our model in different stages. We start by using NUS-48E Corpus [23] to train the SSAN model, since NUS-48E provides the paired speech and singing voice data. For SVC model training, we use a combination of singing voice and speech datasets, including OpenSinger [24], ACE-KiSing [25] and Aishell-3 [26], to train SSANSVC model. For evaluation, we utilize M4Singer [1] and ST-CMDS [27] datasets to evaluate the zero-shot SVC capacity with our proposed model. All audio samples are resampled to 24 kHz and normalized. Then we calculate the volume feature, extract the F0 curve along with the voiced flag for each frame by using DIO, the content feature from the 12th layer by utilizing ContentVec (i.e., $c_{\{\text{Speech, Sing}\}}$ in Figure 2), and the 192-dimensional timbre feature (i.e., $t_{\{\text{Speech, Sing}\}}$ in Figure 2) from Section 2.1. These features are respectively projected to 256 dimensions with a linear neural layer and then concatenated as the conditional input for the decoding stage. We use the pre-trained singing voice vocoder released in the M4singer work, and the mel-spectrograms are computed with 512-point fast Fourier transform, 512-point window size, and 128-point hop size with 80 frequency bins. Mel spectrograms are min-max normalized to the range [-1, 1].

For the SSAN model, we use a Transformer-based encoder with six layers, each containing 8 attention heads, a 512-dimensional feed-forward network, and a dimension of 192 for attention calculation. Dropout is set to 0.1 to prevent overfitting. The batch size is set to 16 and the learning rate is $3e-6$. We collected a validation set comprising both speech and singing data from multiple singers as a validation set for the SSAN model. Specifically, the dataset includes recordings from 4 singers, each contributing about 4 minutes of speech and 2 minutes of singing data. The detailed training process is provided in Algorithm 1. The training process is stopped when the

Table 1: *Objective Evaluations for SVC, where speech $\rightarrow$ singing refers to converting the singing voice M4Singer to target timbres in the ST-CMDS.*

Method	SIM	SingMOS	SHEET-SSQA	CER
Speech $\rightarrow$ Singing				
CoMoSVC	0.448	3.820	3.413	40.43
SSANSVC-stage1	0.508	3.821	3.428	29.2
SSANSVC-stage2	0.506	3.871	3.425	36.56

Table 2: *Objective Evaluations for SVC, where singing $\rightarrow$ singing refers to converting the singing voice from M4Singer to target timbres within the M4Singer dataset, and speech $\rightarrow$ speech refers to converting the speech voice from ST-CMDS to target timbres within the ST-CMDS dataset.*

Method	SIM	SingMOS	SHEET-SSQA	CER
Singing $\rightarrow$ Singing				
CoMoSVC	0.551	3.910	3.781	16.19
SSANSVC-stage1	0.582	3.868	3.742	16.95
SSANSVC-stage2	0.583	3.918	3.841	16.48
Speech $\rightarrow$ Speech				
CoMoSVC	0.596	3.763	3.802	36.10
SSANSVC-stage1	0.568	3.777	3.740	34.55
SSANSVC-stage2	0.598	3.728	3.726	36.52

validation similarity errors do not decrease in five successive epochs with a maximum of 20 epochs.

In this paper, the CoMoSVC method is used as the baseline method for comparison. We compare the performance of the proposed SSANSVC with stage 1 in Figure 2, as well as with stage 2 in Figure 2, respectively. The CoMoSVC is trained with 400k using the timbre feature extracted by ESPnet. The SSANSVC with stage 1 (named SSANSVC-stage1) is trained with 400k iterations using the timbre feature extracted by SSAN and 200k iterations using the timbre feature extracted by ESPnet. The SSANSVC with stage 2 (named SSANSVC-stage2) is trained with 200k iterations using data generated in stage 2, 100k iterations using the timbre feature extracted by SSAN and 200k iterations using the timbre feature extracted by ESPnet. All the models are trained on a single NVIDIA A800 GPU with a batch size of 24, with learning rates of $1e-4$ and $5e-5$ respectively and the optimizer is AdamW [28].

3.2. Evaluation

We evaluate the performance of different methods by subjective and objective metrics. For the subjective test, we invited 10 volunteers to give Mean Opinion Scores (MOS) on naturalness (MOS-n) and timbre similarity (MOS-ts) on the converted audio. We use the 5-point Likert scale for testing with MOS-n (1-bad, 2-poor, 3-fair, 4-good, 5-excellent). And we use the 3-point Liker scale for testing with MOS-ts (1-not match, 2-normal, 3-match). We use VERSA to conduct objective evaluations [29]. Specifically, we include Character Error Rate (CER) obtained by Whisper [30] and speaker SIMilarity (SIM) calculated by the cosine distance between the speaker embeddings of the generated audio and the target timbre audio are used as the objective metrics for SVC evaluation. SingMOS [31] and SHEET-SSQA MOS score [32] are used additionally for naturalness evaluation.

3.3. Zero-shot SVC Performance

We conduct three sets of experiments to evaluate performance using objective metrics: (a) using the ST-CMDS dataset for the target singer and the M4Singer dataset as the source for conver-

Table 3: *Subjective evaluations for SVC of converting the singing voice from M4Singer to target timbres in the ST-CMDS dataset.*

Method	MOS-ts	MOS-n
CoMoSVC	1.597 ± 0.055	3.511 ± 0.065
SSANSVC-stage1	1.618 ± 0.054	3.557 ± 0.062
SSANSVC-stage2	1.627 ± 0.057	3.568 ± 0.066

sion; (b) using the M4Singer dataset for both the target singer and the source for conversion; (c) using the ST-CMDS dataset for both the target singer and the source for conversion. The source audios are all unseen during training.

The objective evaluations for Experiment a are shown in Table 1, while the objective evaluations for Experiments (b) and (c) are presented in Table 2. As illustrated in Table 1, the two SSANSVC models show significant improvements over CoMoSVC in all metrics, especially in terms of SIM. SSANSVC-stage1 and SSANSVC-stage2 perform similar performance in SIM. However, SSANSVC-stage1 and SSANSVC-stage2 have their own strengths and weaknesses in different dimensions: SSANSVC-stage1 performs better in conversion intelligibility, while SSANSVC-stage2 excels in singing quality.

As illustrated in Table 2, in the singing $\rightarrow$ singing task, SSANSVC-stage2 performs best in terms of SIM and audio quality score, despite having a slightly higher CRR than CoMoSVC. In the speech $\rightarrow$ speech task, SSANSVC-stage2 still excels in similarity, while SSANSVC-stage1 outperforms the other two methods in CER. Overall, SSANSVC-stage2 shows the most balanced performance and stands out in terms of CER.

The subjective evaluations of 10 volunteers in Table 3 for the speech $\rightarrow$ singing task reveal that SSANSVC-stage2 performs the best in both timbre similarity and naturalness, demonstrating higher audio quality and naturalness scores. SSANSVC-stage1 follows closely behind, slightly lagging in both metrics, while CoMoSVC shows relatively weaker performance.

4. Conclusion

In this paper, we propose a novel network (SSAN) designed to adapt speaker and singer embeddings. In addition, we design a cycle training strategy for model optimization. This strategy ensures that the network not only learns to convert the voice from one domain (e.g., speech) to another (e.g., singing) but also guarantees that the converted voice maintains consistency with the original speaker’s identity and singing style. Through comprehensive subjective and objective evaluations, we demonstrate the effectiveness of the proposed method.

Though our proposed method achieves better performance, it still requires further improvements. The primary area of enhancement lies in the loss function, where it is important to not only consider mel-loss but also incorporate timbre loss and lyrics recognition loss. By doing so, we can more effectively capture the nuances of both the singer’s timbre and the correct lyrical content during the conversion. Additionally, we plan to explore a more efficient training approach by eliminating the two-stage training method, and opting for a direct one-stage solution. This could simplify the training process and potentially improve the overall performance of the model.

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